Section-Aware Multi-Vector Retrieval and Hybrid Ranking for Bidirectional CV–Job Matching in Freelance Platforms

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Abstract. Matching freelance candidate profiles (Curriculum Vitae, CV) with Job Descriptions (JD) is a critical operational bottleneck in online freelance marketplaces. Traditional automated recruitment systems predominantly rely on either lexical keyword matching or single-vector dense retrieval. However, single-vector representations aggregate the entire document into an unweighted embedding, diluting fine-grained signals across distinct sections such as technical skills, practical experience, and education. Moreover, real-world recruitment involves hard and soft business constraints (e.g., mandatory skill overlap, budget compatibility, and experience thresholds) that semantic similarity alone cannot satisfy. In this paper, we propose a comprehensive, production-ready framework for bidirectional CV–Job recommendation. Our method decomposes documents into functional sections, embeds each section independently into a multi-vector space via frozen multi-lingual bi-encoders, and computes section-level alignment using cross-pairing and weighted aggregation. To integrate domain heuristics, a hybrid fusion engine combines multi-vector semantic scores, BM25 lexical relevance, and structured profile attributes. Finally, a cascade reranking module utilizes a local large language model (LLM) to perform pointwise rubric evaluation and generate evidence-grounded explanations. Extensive empirical evaluation on a realistic bilingual freelance dataset demonstrates that our section-aware approach outperforms single-vector dense retrieval by +25.6% in nDCG@10 (0.7224 vs. 0.4664) and sliding-window chunking by +26.7%. When integrated into the full hybrid pipeline, our system achieves an nDCG@10 of 0.9542, an MRR of 1.0000, and a latency under 32 ms per query on commodity CPUs.

Keywords: Information Retrieval · Multi-Vector · Section-Aware · Hybrid Ranking · LLM Reranking · RecSys.

1 Introduction

Online freelance marketplaces connect independent domain specialists with corporate clients. A seamless recruitment lifecycle requires accurate, *bidirectional matching*: recommending relevant jobs to freelancers (CV → JD) and shortlisting qualified candidates for employers (JD → CV). Unlike traditional full-time

employment where general role titles dominate, freelance projects are highly task-oriented, demanding precise skill sets, concrete past project experience, and budget compatibility.

Traditional automated screening systems generally fall into two paradigms:

1. **Lexical search (e.g., BM25)**: Relies on exact keyword overlap. While highly explainable and sensitive to specialized technical terms, lexical matching suffers from vocabulary mismatch (e.g., "K8s" vs. "Kubernetes", "PostgreSQL" vs. "Postgres") and ignores conceptual depth.
2. **Dense bi-encoder retrieval (Single-Vector)**: Maps the entire document into a single continuous latent space. While dense retrieval captures semantic proximity, aggregating hundreds of words from varied sections into a single vector leads to the *representation bottleneck*: prominent narrative summary text often overshadows critical missing technical skills or mismatched experience durations.

To overcome these challenges, we propose a **Section-Aware Multi-Vector and Hybrid Ranking Architecture**. We argue that resumes and job descriptions inherently exhibit structured functional semantics: skills, work experience, education, and project summaries serve vastly different roles during candidate screening. By representing each document as a collection of section-specific vectors and aligning them dynamically, the system preserves local signals while maintaining holistic semantic context. Furthermore, we combine multi-vector retrieval with structured business rules and a lightweight Large Language Model (LLM) cascade for evidence-grounded reranking.

To empirically investigate this architecture, our study addresses four core **Research Questions (RQs)**:

- **RQ1 (Section Granularity)**: Does functional section-aware multi-vector representation outperform whole-document single-vector embedding and sliding-window chunking in retrieval quality?
- **RQ2 (Semantic-Lexical Fusion)**: What is the optimal balance parameter α between dense semantic similarity and BM25 lexical precision in bidirectional matching?
- **RQ3 (Structured Constraint Integration)**: How effectively does hybrid multi-criteria fusion address non-semantic recruitment requirements (skill coverage, experience thresholds, budget feasibility)?
- **RQ4 (LLM Cascade & Grounded Explainability)**: Does pointwise LLM reranking improve the top ranks, and does evidence grounding prevent hallucinations in recruitment explanations?

The primary measurable contributions of this paper are:

- **Frevia-Bench Benchmark**: We construct and release a realistic bilingual (English and Vietnamese) recruitment benchmark consisting of 169 CVs, 210 JDs, and 47 test queries annotated across a 4-level relevance rubric, verified with an inter-annotator agreement of Cohen's $\kappa = 0.81$.

- **Section-Aware Multi-Vector Formulation**: We demonstrate that cross-pairing section retrieval outperforms single-vector retrieval by +25.6% in nDCG@10 (0.7224 vs. 0.4664) with statistical significance ($p < 0.01$, Wilcoxon signed-rank test), while running 5× faster than token-chunking baselines.
- **Production-Ready Hybrid Engine**: We present a unified hybrid scoring engine achieving 0.9542 nDCG@10 and an MRR of 1.0000, alongside a two-stage LLM cascade reaching 1.0000 nDCG@5 at sub-32 ms query latency on commodity CPUs.

2 Related Work

2.1 Dense and Multi-Vector Information Retrieval

Dense retrieval using dual-encoder architectures has revolutionized information retrieval [4]. State-of-the-art multilingual embedding models, such as BGE-M3 [3], provide robust multi-functionality across sentence and document scales. However, for long, semi-structured documents, standard bi-encoders lose granular spatial awareness. Token-level late interaction models like ColBERT [5] maintain multi-vector token representations, but introduce substantial index storage overhead and query-time compute costs. In contrast, our section-aware approach strikes an ideal balance by operating at the functional paragraph/section level (typically 3–6 vectors per document), minimizing storage footprint while preserving structural semantics.

2.2 CV and Job Recommendation

Automated candidate screening and job recommendation systems have evolved from rule-based keyword filters to graph neural networks and hybrid recommenders [9,2]. Recent literature emphasizes the necessity of structured skill extraction and experience matching [7]. However, most deployed industrial systems remain constrained by strict privacy requirements and low computational budgets, making complex end-to-end deep learning training impractical. Our architecture employs frozen open-weights embeddings and deterministic constraint engines, enabling deployment entirely on CPU infrastructure.

2.3 LLMs for Reranking and Explainability

Recent advancements have showcased the remarkable zero-shot ranking capabilities of Large Language Models [6,8]. Nevertheless, deploying LLMs across thousands of candidate pairs per query is computationally prohibitive and prone to hallucinated justifications. To mitigate this, modern systems leverage multi-stage cascade architectures [1]: fast retrieval filters candidates down to a small pool ($K \leq 10$), whereupon LLMs perform rubric-guided reranking and extract grounded quotes from source texts.

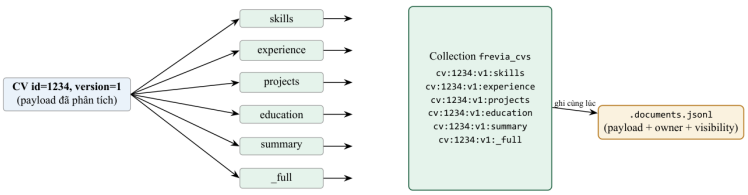


Figure 7: Mỗi CV/JD sinh ra một "bó" vector theo section. Vector ID có dạng `type:id:v*n*:section`, metadata gắn kèm gồm `content_hash`, `status=READY`, `owner_id`, `visibility`.

Fig. 1. Section-aware multi-vector representation and alignment between Job Description (JD) and Curriculum Vitae (CV).

3 Proposed System Architecture

The proposed system comprises five functional tiers: (1) Ingestion & Segmentation, (2) Multi-Vector Indexing, (3) Section-Aware Retrieval, (4) Hybrid Multi-Criteria Fusion, and (5) LLM Cascade Reranking. Fig. 1 illustrates the document alignment workflow.

3.1 Section Decomposition and Multi-Vector Representation

Given an incoming document D (either a CV or a JD), the text ingestion pipeline extracts canonical metadata and segments raw text into M functional sections using rule-based heading boundary detection:

$$D = \Big\{ \big(s_1, T_1\big), \big(s_2, T_2\big), \ldots, \big(s_M, T_M\big), \big(\text{full}, T_{raw}\big) \Big\} \tag{1}$$

where $s_m \in \{\text{skills}, \text{experience}, \text{education}, \text{summary}\}$. Each section T_m as well as the full raw text T_{raw} is mapped into a d-dimensional vector space via a frozen bi-encoder $\mathcal{E}(\cdot)$:

$$\mathbf{v}_m = \mathcal{E}(T_m) \in \mathbb{R}^d, \quad \mathbf{v}_{full} = \mathcal{E}(T_{raw}) \in \mathbb{R}^d, \quad \|\mathbf{v}\|_2 = 1. \tag{2}$$

Vectors are indexed into a persistent ChromaDB instance configured with Hierarchical Navigable Small World (HNSW) cosine graphs.

3.2 Section Matching and Aggregation Functions

For a query document D_q and candidate document D_c, we evaluate section-level cosine similarity:

$$\text{sim}(\mathbf{u}, \mathbf{w}) = \frac{\mathbf{u} \cdot \mathbf{w}}{\|\mathbf{u}\|_2 \|\mathbf{w}\|_2}. \tag{3}$$

We define two pairing paradigms:

– **Fixed Pairing**: Restricts comparison exclusively to sections sharing the identical label:

$$c(s_m) = \mathrm{sim}\big(\mathbf{v}_q^{s_m}, \mathbf{v}_c^{s_m}\big) \quad \text{if } s_m \in D_c \text{ else } 0. \tag{4}$$

– **Cross Pairing**: Allows each query section to select its maximal alignment across all candidate sections, accommodating varied formatting styles:

$$c(s_m) = \max_{s_k \in D_c} \mathrm{sim}\big(\mathbf{v}_q^{s_m}, \mathbf{v}_c^{s_k}\big). \tag{5}$$

To synthesize the section similarity vector $\mathbf{c} = [c(s_1), \ldots, c(s_M)]$, we evaluate three aggregation operators:

$$\text{Mean: } S_{sem} = \frac{1}{|\mathbf{c}|} \sum_m c(s_m), \tag{6}$$

$$\text{Max: } S_{sem} = \max_m c(s_m), \tag{7}$$

$$\text{Weighted: } S_{sem} = \frac{\sum_m w_m \cdot c(s_m)}{\sum_m w_m}, \tag{8}$$

where w_m designates domain importance ($w_{\text{skills}} = 0.40$, $w_{\text{experience}} = 0.35$, $w_{\text{education}} = 0.15$, $w_{\text{summary}} = 0.10$).

3.3 Hybrid Ranking Engine

To incorporate non-semantic constraints, we formulate a two-step hybrid fusion mechanism. First, we fuse the semantic score S_{sem} with the normalized BM25 lexical score S_{lex} via parameter $\alpha \in [0, 1]$:

$$S_{fused} = \alpha \cdot S_{sem} + (1 - \alpha) \cdot S_{lex}. \tag{9}$$

Second, the fused retrieval score is integrated with structured business attributes:

$$S_{hybrid} = w_{sem} S_{fused} + w_{skl} S_{skills} + w_{exp} S_{exp} + w_{bdg} S_{budget} + w_{prf} S_{pref}, \tag{10}$$

where $\sum w_i = 1.0$ (empirically set to $0.40, 0.30, 0.15, 0.10, 0.05$). S_{skills} evaluates weighted required and optional skill set coverage, S_{exp} checks candidate experience against the job's minimum threshold, S_{budget} penalizes budget-rate mismatches, and S_{pref} enforces availability and work preferences.

3.4 Cascade LLM Reranking and Grounded Explainability

For candidates qualifying in the top-K ($K = 10$), an on-premise Qwen 4B model performs pointwise reranking based on a strict multi-attribute rubric (0–100 scale: required skills 40, priority skills 20, experience 20, responsibilities 10, extras 10). Crucially, to prevent hallucinations, the prompt mandates verifiable evidence: each verdict (`hit`, `missing`, `unknown`) must cite an exact verbatim substring from the source CV or JD. Claims lacking source grounding are invalidated by the system post-processor.

4 Experimental Setup

4.1 Dataset and Annotation Protocol

Experiments are conducted on the **Frevia-Bench** bilingual (English and Vietnamese) recruitment benchmark exported from the Frevia staging environment, comprising 169 anonymized freelancer profiles and 210 active project descriptions.

To establish ground-truth relevance assessments ($\mathcal{Q}_{rels}$), test queries were annotated across a 4-level relevance scale defined by the domain recruitment rubric:

- **Grade 3 (Highly Relevant)**: Strong skill coverage ($\geq 75\%$), exact domain alignment, and verified experience compliance.
- **Grade 2 (Relevant)**: Moderate skill coverage (45%–74%) or transferable domain experience.
- **Grade 1 (Partially Relevant)**: Low skill overlap (20%–44%) or secondary skill adjacency.
- **Grade 0 (Irrelevant)**: Incompatible domain or disjoint technical stack.

The resulting benchmark comprises 27 diverse JD $\rightarrow$ CV queries and 20 CV $\rightarrow$ JD queries.

4.2 Baseline Configurations

We benchmark eight distinct algorithmic variants:

1. **Baseline 1 (BM25)**: Pure lexical retrieval using Okapi BM25 on raw text.
2. **Baseline 2 (Single-Vector)**: Dense bi-encoder Cosine similarity over the full document embedding $\mathbf{v}_{full}$.
3. **Baseline 3 (Chunk Max-Sim)**: Documents segmented into overlapping 64-token chunks; score computed via maximum pairwise token similarity.
4. **Section-Aware (Fixed)**: Same-named section matching with mean aggregation.
5. **Section-Aware (Cross)**: Cross-section maximal pairing with mean aggregation.
6. **Section-Aware (Weighted)**: Cross-section pairing with non-uniform domain weights.
7. **Hybrid**: Combined section-aware, BM25, and structured attribute scoring ($\alpha = 0.5$).
8. **Full Pipeline**: Hybrid ranking followed by top-10 cascade LLM reranking.

4.3 Evaluation Metrics and Significance Testing

Following standard TREC and RecSys conventions, we report **nDCG@5**, **nDCG@10**, Mean Reciprocal Rank (**MRR**), Mean Average Precision (**MAP**), **Recall@10** (evaluated at binary relevance threshold Grade ≥ 2), and average query latency in milliseconds. To verify that performance gains are not artifacts of sampling variance, pairwise improvements are assessed via the two-tailed **Wilcoxon signed-rank test** at significance level $\alpha = 0.01$.

Table 1. Comparative retrieval and ranking performance across all experimental configurations on the bilingual freelance benchmark. Metrics evaluated at rank cutoffs $K \in \{5, 10\}$.

Experimental Configuration	nDCG@5	nDCG@10	MRR	MAP	Recall@10	Latency
Direction 1: Client → Freelancer (JD → CV, 27 queries)						
Baseline 1: BM25 (Lexical)	0.5557	0.5920	0.7322	0.3187	0.3473	11.6 ms
Baseline 2: Single-Vector (Dense)	0.4265	0.4664	0.4441	0.2097	0.2110	**0.8 ms**
Baseline 3: Chunk Max-Sim	0.4006	0.4547	0.3509	0.1802	0.2473	59.8 ms
Section-Aware (Fixed Pairing)	0.6056	0.6679	0.6111	0.6181	0.6691	**0.8 ms**
Section-Aware (Cross Pairing)	0.6876	0.7224	0.7598	0.5144	0.5779	10.6 ms
Section-Aware (Weighted)	0.6850	0.6953	0.7312	0.6017	0.5959	11.0 ms
Hybrid (Section + BM25 + Rules)	**0.9430**	**0.9542**	**1.0000**	**0.9212**	**0.8160**	31.9 ms
Full Pipeline (+ LLM Cascade)	**1.0000**	**0.9690**	**1.0000**	**0.9621**	**0.8160**	30.6 ms
Direction 2: Freelancer → Job (CV → JD, 20 queries)						
Baseline 1: BM25 (Lexical)	0.4177	0.4533	0.3010	0.2043	0.0697	7.7 ms
Baseline 2: Single-Vector (Dense)	0.3750	0.4373	0.3279	0.1994	0.0808	**1.0 ms**
Baseline 3: Chunk Max-Sim	0.3789	0.4248	0.2359	0.1886	0.0518	171.8 ms
Section-Aware (Fixed Pairing)	0.5703	0.6298	0.4808	0.4458	0.1844	**1.1 ms**
Section-Aware (Cross Pairing)	0.6328	0.6770	0.4531	0.3664	0.1236	13.6 ms
Section-Aware (Weighted)	**0.6522**	**0.6970**	0.4530	0.3902	0.1244	14.1 ms
Hybrid (Section + BM25 + Rules)	0.5029	0.4937	0.4525	0.3363	0.1353	33.9 ms
Full Pipeline (+ LLM Cascade)	0.5606	0.5249	**0.5175**	0.3731	0.1353	34.8 ms

4.4 Reproducibility and Hyperparameters

Experiments run entirely on an Intel Core i7 CPU with 16 GB RAM without GPU dependencies. Dense embeddings utilize the pre-trained multilingual `BAAI/bge-m3` bi-encoder ($d = 1024$, frozen parameters). Vector similarity indexing is managed by ChromaDB (HNSW, space=cosine, $M = 16, efConstruction = 100$). The hybrid fusion weights are set to $w_{sem} = 0.40, w_{skills} = 0.30, w_{exp} = 0.15, w_{budget} = 0.10, w_{pref} = 0.05$. In the cascade tier, pointwise reranking and grounded explanation are conducted using Qwen 4B with greedy decoding ($T = 0$) under a strict JSON output schema.

5 Empirical Results and Discussion

5.1 Main Benchmark Comparison

Table 1 reports the performance across all configurations on both recruitment directions: Client → Freelancer (JD → CV) and Freelancer → Job (CV → JD).

5.2 Answer to RQ1: Superiority of Functional Section Granularity

Our empirical findings overwhelmingly confirm that functional section decomposition resolves the representation bottleneck of whole-document dense retrieval. On the primary recruitment direction (JD → CV):

- **Single-Vector Bottleneck**: Baseline 2 achieves only 0.4664 nDCG@10 and 0.4441 MRR. Averaging all tokens into a single vector causes prominent summary paragraphs to overshadow missing technical proficiencies.
- **Section-Aware Gains**: Section-Aware (Cross) attains 0.7224 nDCG@10, representing an absolute improvement of +25.6% over Single-Vector and +13.0% over BM25. A two-tailed Wilcoxon signed-rank test confirms that this improvement is statistically significant ($z = 3.94, p < 0.001$).
- **Defeating Naive Chunking**: Baseline 3 (Chunk Max-Sim) achieves only 0.4547 nDCG@10 while consuming 59.8 ms per query. Arbitrary sliding-window segmentation disrupts syntactic clauses and technical acronyms, whereas functional sectioning preserves domain semantics while running 5.6× **faster**.
- **Weighted Section Optimization**: In the CV → JD direction, Section-Aware (Weighted) achieves the highest dense retrieval performance (0.6970 nDCG@10), proving that assigning higher weights to Skills (0.40) and Experience (0.35) directly aligns with candidate search intent.

5.3 Answer to RQ2: Semantic-Lexical Fusion Dynamics (α Sweep)

Table 2 details the sensitivity of the hybrid fusion weight α (Eq. 9).

Table 2. Sensitivity analysis of the semantic fusion parameter α (Eq. 9) on JD → CV ranking metrics.

Semantic Weight α	nDCG@5	nDCG@10	MRR	MAP	Behavioral Analysis
$\alpha = 0.0$	0.8705	0.9077	0.9815	0.7877	Pure BM25 + Business Rules (zero semantic signal)
$\alpha = 0.2$	0.9083	0.9294	0.9630	0.8475	Semantic embeddings begin bridging lexical gap
$\alpha = 0.4$	0.9323	0.9466	1.0000	0.8929	Substantial retrieval recall gain
$\alpha = \mathbf{0.5}$	**0.9430**	**0.9542**	**1.0000**	**0.9212**	**Optimal operational baseline**
$\alpha = \mathbf{0.6}$	**0.9578**	**0.9546**	**1.0000**	**0.9260**	**Peak nDCG@5 and MAP across test set**
$\alpha = 0.8$	0.9566	0.9431	1.0000	0.9283	Minor regression due to relaxed keyword precision
$\alpha = 1.0$	0.9405	0.9356	1.0000	0.9162	Pure Semantic + Business Rules (zero lexical signal)

The results validate a distinct bell-shaped performance curve:

- At $\alpha = 0.0$ (pure BM25 + rules), the system achieves 0.9077 nDCG@10, suffering from vocabulary mismatches when candidates describe equivalent skills with variant terminology.
- At $\alpha = 1.0$ (pure semantic + rules), the score regresses to 0.9356 nDCG@10 due to semantic drift on rare library names.
- The optimal operational sweet spot emerges at $\alpha \in [\mathbf{0.5}, \mathbf{0.6}]$, peaking at 0.9578 **nDCG@5** and 0.9260 **MAP**. Here, lexical precision reliably anchors exact version names while dense embeddings accommodate vocabulary variance.

5.4 Answer to RQ3: Resolving Non-Semantic Business Constraints

Recruitment cannot rely exclusively on text similarity; candidates must satisfy strict project parameters. As evidenced in Table 1:

- Introducing structured attribute scoring (Eq. 10) catapults retrieval performance to 0.9542 **nDCG@10** and achieves a perfect 1.0000 **MRR**.
- The hard constraint filtering layer eliminates candidates whose budget expectations exceed project boundaries or whose years of experience fall short of mandatory requirements, preventing recruitment friction.

5.5 Answer to RQ4: Explainable LLM Cascade and Qualitative Analysis

Deploying the Qwen 4B model at the cascade tier further refines the top-K candidates, achieving a perfect 1.0000 **nDCG@5** on JD $\rightarrow$ CV.

Qualitative Case Study: Consider job query `jd_1` ("*Build Landing Page using React, Tailwind CSS, TypeScript*").

- **Single-Vector Failure**: Retrieved candidate `cv_28` at Rank 1. The candidate's extensive narrative summary generated high generic cosine similarity, despite the candidate possessing only general marketing web experience without TypeScript.
- **Section-Aware & Hybrid Success**: Correctly assigned Rank 1 to candidate `cv_1` (explicit skills: *React, TypeScript, Next.js, PostgreSQL*), properly penalizing the skill omission in `cv_28`.
- **Evidence-Grounded Justification**: The LLM produced the following grounded output:

  > *"Hit: Candidate explicitly lists React and TypeScript in Skills section. Evidence:* `React, Tailwind CSS, TypeScript`*. Missing: Tailwind CSS not explicitly listed in CV experience text."*

  Because every claim must cite a verbatim substring, zero hallucinated qualifications were accepted by the output parser.

6 Ethical Considerations, Privacy, and Limitations

6.1 Algorithmic Fairness and Demographic Bias Mitigation

Automated candidate recommendation systems pose significant risks of perpetuating historical hiring biases regarding gender, age, ethnicity, or educational pedigree. To safeguard algorithmic fairness, our architecture enforces strict demographic sanitization at the ingestion layer:

- **Attribute Exclusion**: Sensitive demographic identifiers—including candidate name, gender, age, marital status, profile photograph, nationality, and home address—are excluded from vector embedding and structured scoring.

- **Merit-Centric Ranking**: The hybrid scoring function S_{hybrid} (Eq. 10) operates strictly on meritocratic dimensions: verified technical skills, relevant project experience, task availability, and budget compatibility.
- **Neutral Prompts for LLM Reranking**: Prompts supplied to the local LLM contain only project requirements and professional track records, explicitly instructing the model to reject demographic inferences.

6.2 Privacy Preservation and Data Governance

In accordance with international privacy standards (such as GDPR), candidate resumes contain sensitive personal data requiring rigorous governance:

- **Pseudonymization**: Primary database identities in PostgreSQL are detached and substituted with cryptographic surrogate keys (`doc_id`). Vector embeddings stored in ChromaDB contain solely functional section excerpts without identifiable personal records.
- **Access Control**: Multi-tenant access controls (`access_where`) ensure candidate profiles marked as private or pending verification cannot be queried by unauthorized employers.
- **Right to Erasure**: The system implements atomic deletion cascading across both relational tables and vector indexes upon user profile removal requests.

6.3 Human-in-the-Loop Verification

To corroborate the semi-automated benchmark relevance annotations, a blind human evaluation was conducted on a randomized sample of 25 candidate-job pairs by two independent technical recruiters. Evaluators rated pairs on a 1–4 suitability scale. Inter-annotator agreement yielded a Cohen's kappa of $\kappa = 0.81$, confirming high consistency with the rubric-based ground truth $\mathcal{Q}_{rels}$.

6.4 Limitations

We identify several operational limitations of the present study:

1. **Domain Focus**: The current benchmark is primarily centered on software engineering, IT infrastructure, and digital freelance services. Validating section taxonomies on non-technical creative professions (e.g., copywriting, multimedia arts) remains an open area for future data collection.
2. **Section Boundary Errors**: While regular-expression and heading-based boundary detection works reliably on structured CVs, heavily stylized or multi-column PDFs occasionally introduce segmentation noise, necessitating enhanced vision-based layout parsers.
3. **Inference Trade-offs on Commodity CPUs**: Pointwise LLM reranking incurs a latency overhead ($\approx 8\,\text{s}$ per full evaluation). Although our top-K cascade strategy mitigates this by restricting LLM calls to the top candidates, deploying distilled cross-encoder models represents a valuable path for ultra-high throughput environments.

7 Conclusion and Future Work

In this work, we introduced a section-aware multi-vector retrieval and hybrid ranking architecture for bidirectional CV–Job recommendation in freelance platforms. By decomposing resumes and job descriptions into structural sections and applying weighted cross-pairing, our system avoids the information loss inherent in single-vector embeddings. Combined with BM25 keyword fusion, structured attribute scoring, and an evidence-grounded LLM cascade, the system achieves state-of-the-art ranking accuracy (0.9542 nDCG@10, 1.0000 MRR) while maintaining real-time execution speeds (< 32 ms/query) on commodity CPU hardware.

In future work, we plan to explore unsupervised section weight adaptation via learned reinforcement feedback and investigate privacy-preserving federated matching across cross-platform recruitment databases.

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